

R1

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1 New Code / Dev Stuff

1.1 Python Terminal Script for Filtering

Due to the fact that we need to filter multiple simulations using very similar methods, I decided to make an actual script instead of just using a notebook repeatably. The largest upside is that this can easily be run on a compute node, which makes filtering larger sims easier and faster. Having a single main filtering script also helps make sure any results are reproducible.

Takes a user input via the CLI (see below), filters for all **BH_Sol** systems, and returns the population file with said systems. Additionally, it returns a .csv of just the **BH_Sol** rows. While running, it provides feedback via the terminal stating how many **BH_MS** systems it found, as well as how many are the expected S1:S2 configuration. If any are S2:S1, it prints a warning as well as the total count.

Need to further refine it, but there's also a default SBATCH script to help expedite the submission process.

Usage Example

```
python BH-Sol_Filter.py /Data/1e+00_Zsun_population.h5 -o ./testResults
```

Arguments

Argument	Use	Ex value
-o	output location	./results
-outputName	name for the output files ¹	BH_Sol_Systems
-overwrite	Whether to overwrite existing results, T/F bool. Defaults to F	True
--maxMass	Max mass used when filtering, defaults to $3M_{\odot}$	3.5
--maxPeriod	Max period used when filtering, defaults to 3.5 days	4

Can be found at https://github.com/PiersonLip/stellarBHs/blob/main/FilteringScripts/BH-Sol_Filter.py

¹If a name is not specified it will use the original filename, appending a specifier to the front

2 New Research

2.1 Pop 1

When filtering via notebook and script, both returned 6 systems that had $p_{orb} < 3.5, M_{\odot,2} < 3, t < 10e9$

These systems all evolved via the same channel, ZAMS_oCE1_CC1_oRL02_CC2. Via inspection of the history df and the oneline, it is evident that the systems evolve detached, then into post-MS **unstable** RLO via $S1 \rightarrow S2$. This period decrease leads to common envelope, which then *greatly* decreases the orbital period. The envelope is then ejected, leaving a depleted stripped He core². The core then collapses in a couple of thousand years, leaving the system in a BH-MS state with low p_{orb} /separation.

These systems then evolve in a semi-peculiar way, where it is highly likely that there will be RLO from S2 onto S1, and because $S1 > S2$, this transfer is stable, leading to *very* prolonged stable RLO (seemingly on the scale of $> 10Gyr$), which leaves S2 at low masses ($< .08M_{\odot}$).

Interestingly, the initial period for all the systems is close to $4000d$, with mean of 4022 and a std of only 151. However, a larger sample size is needed to confirm this.

Additionally, the initial masses of S1 and S2 appear to be quite constrained as well.

	orbital_period.i		S1_mass.i		S2_mass.i
count	6.000000	count	6.000000	count	6.000000
mean	4022.716197	mean	17.515761	mean	2.296788
std	151.522384	std	0.520269	std	0.320702
min	3873.118288	min	16.815111	min	1.797809
25%	3908.300983	25%	17.190036	25%	2.120753
50%	3978.305326	50%	17.510176	50%	2.353828
75%	4124.397101	75%	17.818155	75%	2.547210
max	4248.507680	max	18.254965	max	2.620529

2.2 Pop 2

I ran a second grid including an eccentricity distro (Ps and Qs), as well as slightly constraining the initial conditions ($12 < M_1 < 100, .1 < M_2 < 10$)³, hoping to confirm some of the init conditions.

	orbital_period.i		S1_mass.i		S2_mass.i
count	12.000000	count	12.000000	count	12.000000
mean	4136.066721	mean	2.240382	mean	17.357478
std	177.676729	std	0.412075	std	1.073496
min	3844.570082	min	1.551266	min	15.506937
25%	4024.214908	25%	1.903135	25%	16.714083
50%	4119.397535	50%	2.300024	50%	16.998229
75%	4273.345845	75%	2.579111	75%	18.162633
max	4370.857404	max	2.729828	max	19.435080

²I need to read a lot more about CE evo, but my intuition feels weird about the fact that just the envelope of *one* of the stars is ejected

³This is very constrained very early, but I wanted to try something while waiting for the 10 million sim

2.3 Full P_{orb}

When filtering population 2 with a max $p_{orb} = 300\text{days}$ we find 46 BH-Sol systems, all of these (except one!) evolve via the same CE channel. This suggests that there isn't a unique identity placed on BH-Sol systems with $p_{orb} \leq 3\text{days}$ ⁴, and thus it makes more sense to investigate the population as a whole. Note that there is one system which does *not* do RLO before CC1, and only reaches a solar-like mass after RLO2 (upon which it also fails), so I'm excluding it for now. Likely important to keep in mind though.

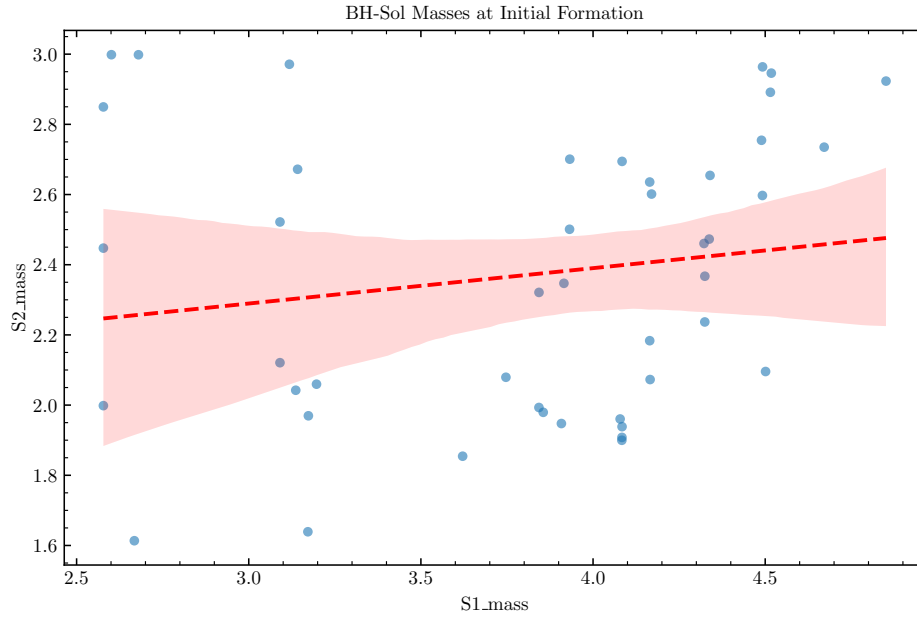


Figure 1: Masses at initial BH-Sol state. i.e., BH formation/S1 collapse

⁴This orbital period distribution is possibly due to natal kick velocities, and is suggested by the literature, but I could not find a strong link in the limited data.

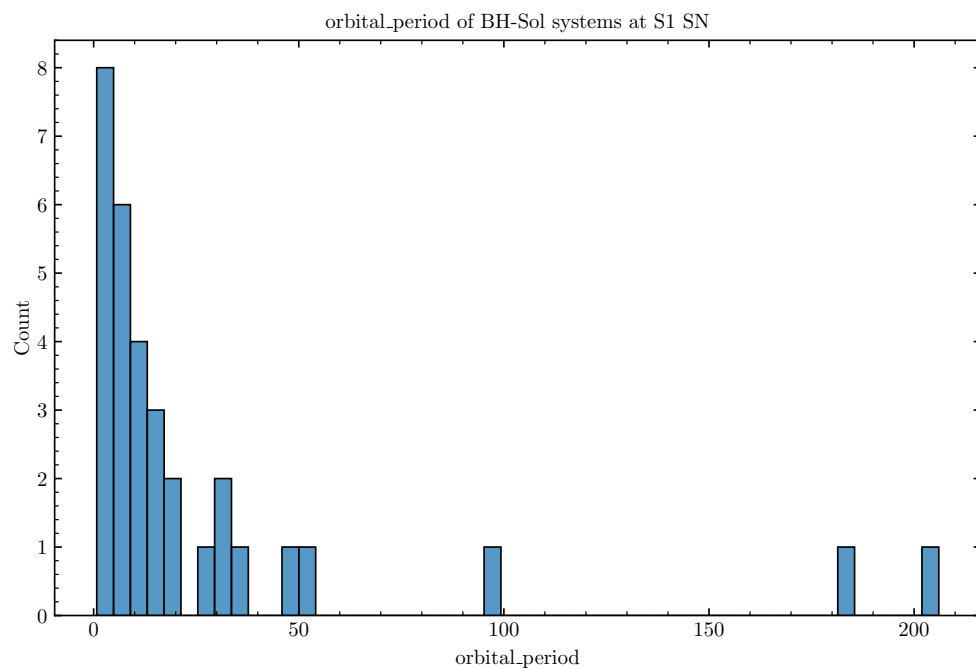


Figure 2: Orbital Period of BH-Sol Systems at init onset.

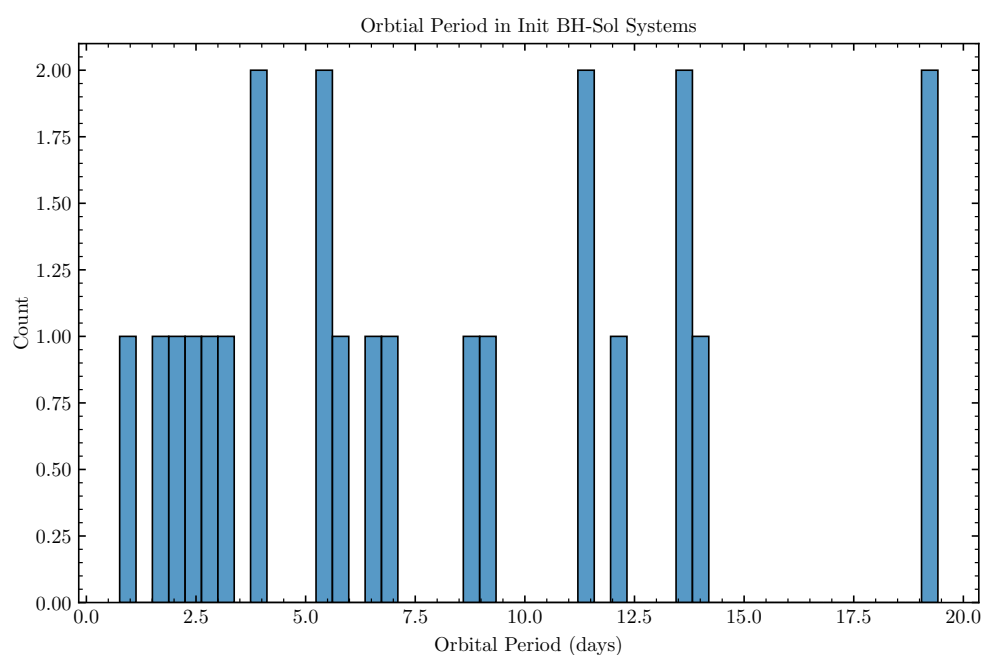


Figure 3: Crop of orbital Period of BH-Sol Systems at init onset.

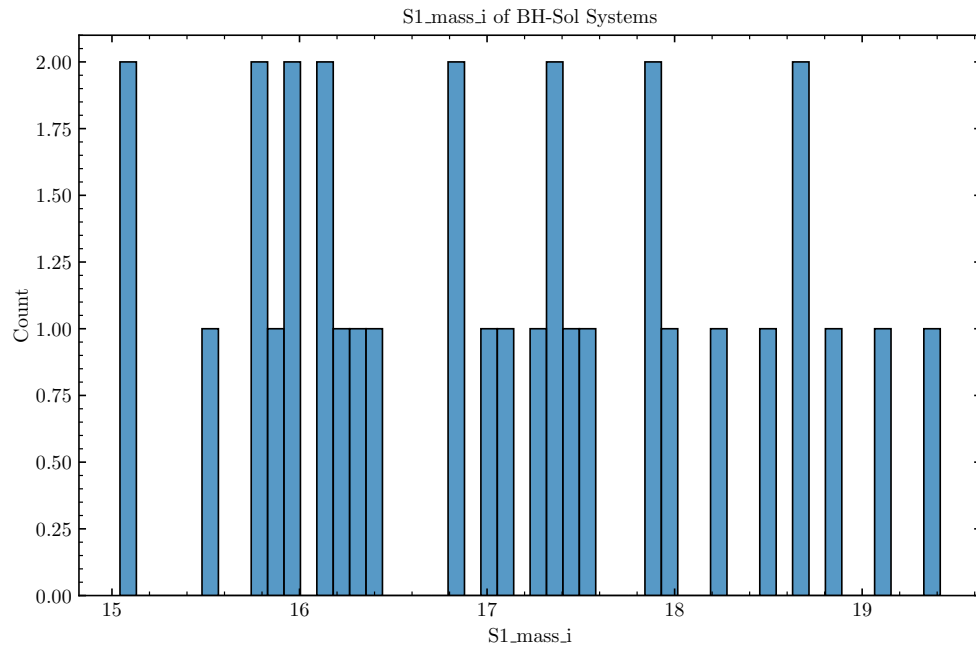


Figure 4: S1 mass initial distribution.

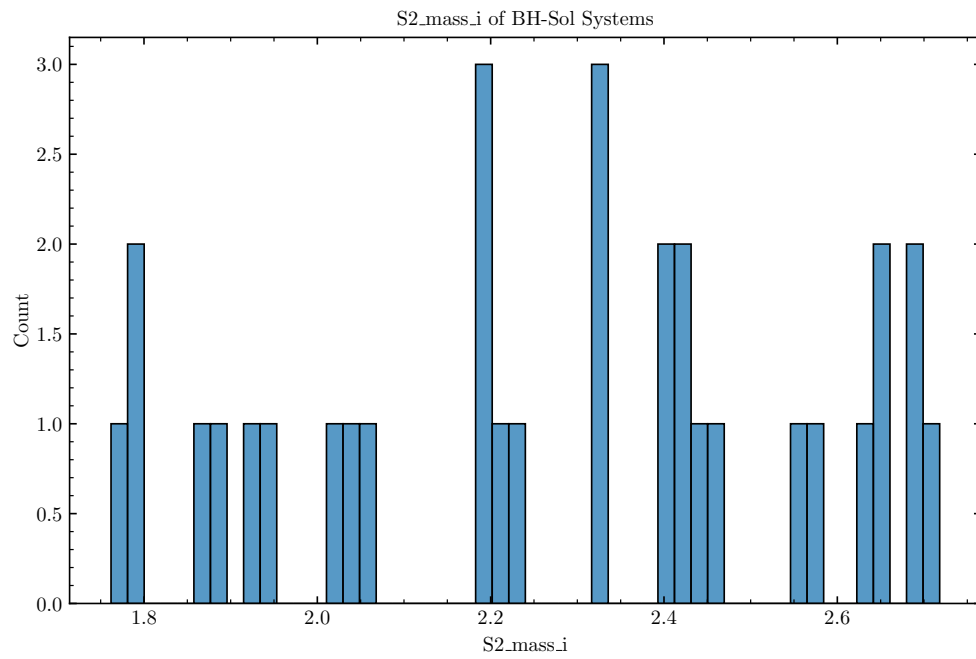


Figure 5: S2 mass initial distribution.

2.4 Strong Initial Masses Correlation

When plotting the individual masses of BH-Sol progenitors from Pop 2 there appeared to be a strong correlation between the two. We see this in figure 6.

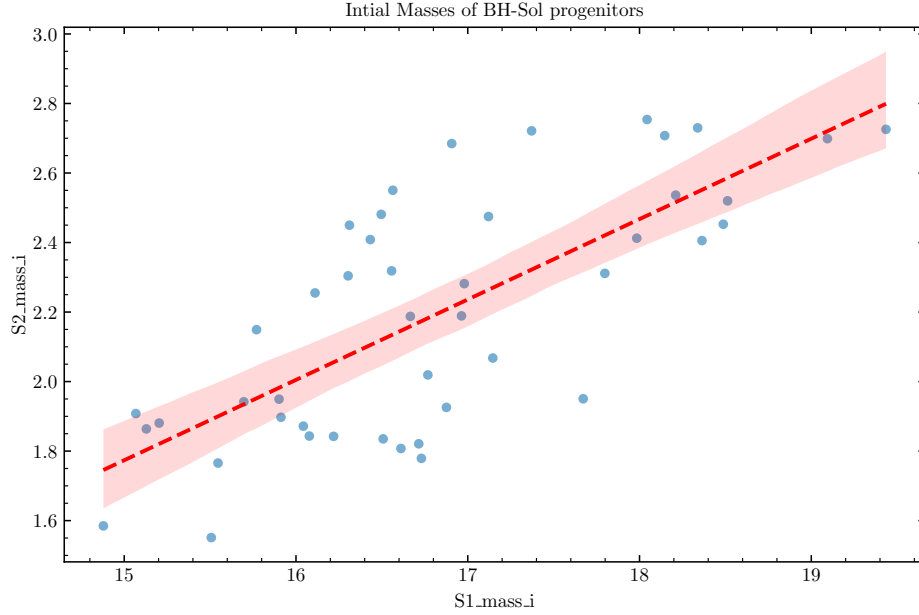


Figure 6: Initial ZAMS masses of simulation two, with mass ratios from Moe17PsandQs. *Note that two failed systems have been excluded*

Initially, I figured this was a biasing from the mass ratio scheme in the simulation being passed through, so I ran another grid with a `secondary_mass_scheme = 'flat_mass_ratio'`. While this would also introduce biasing, I was expecting it to be largely different enough from Moe, and thus could tell if it was simply an artifact or a real correlation. As we can see from figure 7 and 8, this correlation seems to be independent of the initial mass ratio used, suggesting some type of underlying phenomena⁵.

⁵This very likely could just be a survivability configuration of CE evolution

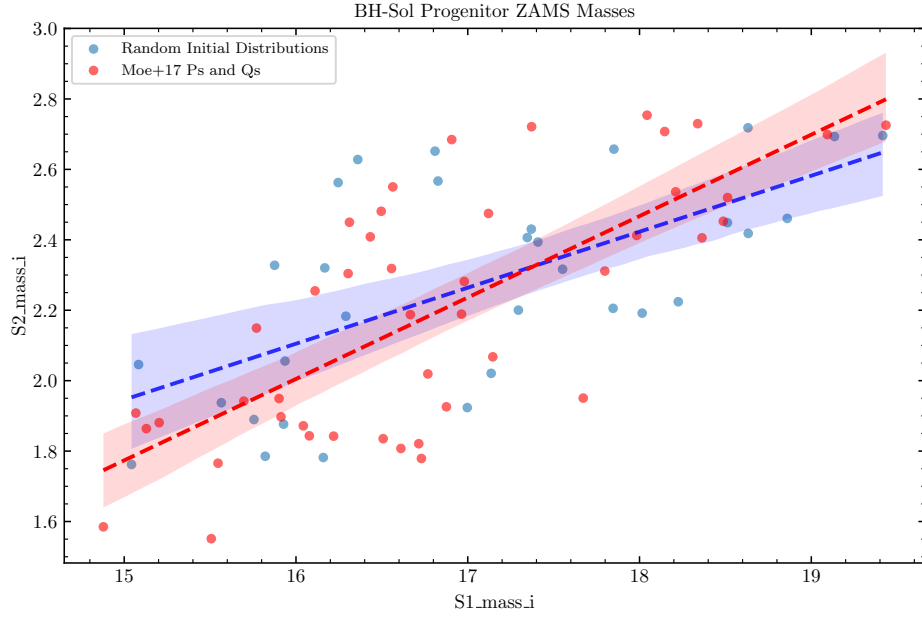


Figure 7: Initial BH-Sol ZAMS masses of simulation two (using Moe17PsandQs) and three (flat mass ratio). *Note that 3 failed systems have been excluded.*

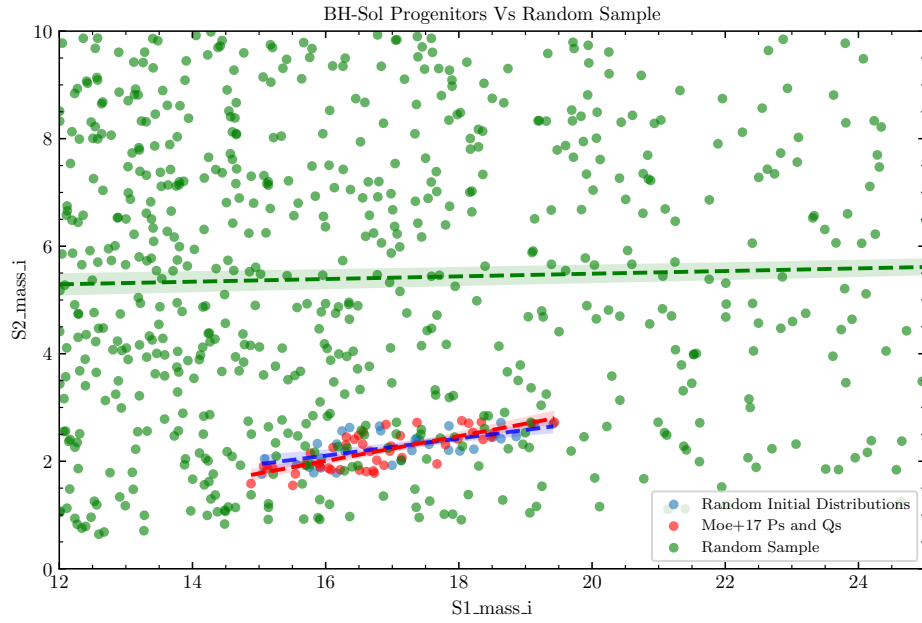


Figure 8: Initial BH-Sol ZAMS masses of simulation two and three, vs random selection of simulation 3. *Note that 3 failed BH-Sol systems have been excluded.*

3 Questions

- Tried running a grid on POSYDON V2.2, seems to have just completely failed from trying to reference itself before it was fully initialized. Entire log was filled with ... After updating to V2.3 it worked fine.

```
Traceback (most recent call last):
File "/gpfs/projects/b1095/bku2126/UCXBInvestigation/POSYDON/posydon/popsyn/binarypopulation.py", 1
    binary.evolve()
File "/gpfs/projects/b1095/bku2126/UCXBInvestigation/POSYDON/posydon/binary_evol/binarystar.py", 1
    self.run_step()
File "/gpfs/projects/b1095/bku2126/UCXBInvestigation/POSYDON/posydon/binary_evol/binarystar.py", 1
    next_step(self)
File "/gpfs/projects/b1095/bku2126/UCXBInvestigation/POSYDON/posydon/binary_evol/DT/step_merged.py
    super().__call__(binary)
File "/gpfs/projects/b1095/bku2126/UCXBInvestigation/POSYDON/posydon/binary_evol/DT/step_isolated.py
    super().__call__(binary)
File "/gpfs/projects/b1095/bku2126/UCXBInvestigation/POSYDON/posydon/binary_evol/DT/step_detached.py
    self.update_after_evo(t, binary, primary, secondary)
File "/gpfs/projects/b1095/bku2126/UCXBInvestigation/POSYDON/posydon/binary_evol/DT/step_detached.py
    current = history[-1]
                ~~~~~~
IndexError: index -1 is out of bounds for axis 0 with size 0
```

and from what I can tell, this happened to nearly all the simmed binaries

- Should we include an initial eccentricity distribution? I imagine it would impact formation with unstable RLO and CE.
- Does it make sense to still be filtering for $p_{orb} < 3\text{days}$? The simulations seem to be suggesting that *all* BH-Sol systems evolve through CE, and that the main thing effecting p_{orb} post CE is the kick strength of the BH. So, maybe we should be initially filtering for all BH-Sol, and if we want the formation rate of specifically low p_{orb} ones, just take the subpop.
- Cool if I present an assortment of general POSYDON things I've worked on at astrofest?
- Any papers on the specifics of CE, and possibly survival rate?
- Also, any possibility of the project going into the summer?